

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Video Projector
Manufacturer: Christie
Firmware Version: N/A
Model(s): D13WU-HS, D13HD-HS

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.04.0000-b007	2.01.0001-b001	1.2.3

Version History

Module Version	Date	Notes
1_0_2_0	6/15/2017	Fixed Video Mute status.
1_0_1_0	2/28/2017	Added Brightness and Contrast.
1_0_0_0	1/12/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example:
InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example:
InterfaceName.connectionCounter = 5

Control Commands, States, and Qualifiers:

AspectRatio ¹	'Auto'	'Native'	'4:3'
	'Letterbox'	'Full Size'	'Full Width'
	'Full Height'	'Custom'	'3D Mode'
AutoImage	'Normal'	'Wide'	
Brightness	1 to 100 in steps of 1		
Contrast	1 to 100 in steps of 1		
Freeze	'On'	'Off'	
Input	'VGA'	'BNC'	'HDMI 1'
	'HDMI 2'	'DVI-D'	'DisplayPort'
	'3G-SDI'	'HDBaseT'	'CVBS'
	'Presenter'	'Card Reader'	'Mini USB'
Keypad	'0' – '9'		
LampMode	'Constant Power'	'Constant Intensity'	'ECO 1'
	'ECO 2'	'Rental Mode'	
MenuNavigation	'Up'	'Down'	'Left'
	'Right'	'Enter'	'Exit'
	'Menu'		
OnScreenDisplay	'On'	'Off'	
PIP	'On'	'Off'	
PIPInput	'VGA'	'BNC'	'HDMI 1'
	'HDMI 2'	'DVI-D'	'DisplayPort'
	'3G-SDI'	'HDBaseT'	'CVBS'
	'Presenter'	'Card Reader'	'Mini USB'
PIPPosition	'Left Vertical Center'	'Top Center'	'Right Vertical Center'
	'Bottom Center'	'Bottom Right'	'Bottom Left'
	'Top Left'	'Top Right'	
PIPSize	'Small'	'Medium'	'Large'
PIPSwap	None		
Power	'On'	'Off'	
VideoMute	'On'	'Off'	

¹ Normal Mode supports all 4:3 input sources. Wide Mode supports all 16:9 input sources and most 4:3 sources.

Status Available:

AspectRatio	'Auto'	'Native'	'4:3'
	'Letterbox'	'Full Size'	'Full Width'
	'Full Height'	'Custom'	'3D Mode'
Brightness	1 to 100 in steps of 1		
Contrast	1 to 100 in steps of 1		
ConnectionStatus	'Connected'	'Disconnected'	
Freeze	'On'	'Off'	
Input	'VGA'	'BNC'	'HDMI 1'
	'HDMI 2'	'DVI-D'	'DisplayPort'
	'3G-SDI'	'HDBaseT'	'CVBS'
	'Presenter'	'Card Reader'	'Mini USB'
LampMode	'Constant Power'	'Constant Intensity'	'ECO 1'
	'ECO 2'	'Rental Mode'	
LampUsage	Hours		
OnScreenDisplay	'On'	'Off'	
OperationHours	Hours		
PIP	'On'	'Off'	
PIPInput	'VGA'	'BNC'	'HDMI 1'
	'HDMI 2'	'DVI-D'	'DisplayPort'
	'3G-SDI'	'HDBaseT'	'CVBS'
	'Presenter'	'Card Reader'	'Mini USB'
PIPPosition	'Left Vertical Center'	'Top Center'	'Right Vertical Center'
	'Bottom Center'	'Bottom Right'	'Bottom Left'
	'Top Left'	'Top Right'	
PIPSize	'Small'	'Medium'	'Large'
Power	'On'	'Off'	
VideoMute	'On'	'Off'	

Cable and Adapter Requirements:

Captive Screw to Female DB9 RS-232 Serial Cable

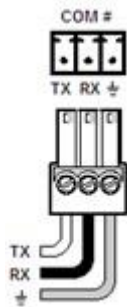
Notes for the Device:

Serial communication:

Port Type: RS-232
Baud Rate: 115200
Data Bits: 8

Parity: None
Stop Bits: One
Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



General Notes:

Network communication:

When configuring the Ethernet interface, be sure the device settings match that of the GS interface settings.

Port Type:	Ethernet
Default Port:	3002
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	No

Ethernet Module Configuration Description:

Please refer to user manual for settings and changes to the network communication

Notes for the Device:

Appendix A. Set Commands

Aspect Ratio 3D Mode	(SZP 8)
Aspect Ratio 4:3	(SZP 2)
Aspect Ratio Auto	(SZP 0)
Aspect Ratio Custom	(SZP 7)
Aspect Ratio Full Height	(SZP 6)
Aspect Ratio Full Size	(SZP 4)
Aspect Ratio Full Width	(SZP 5)
Aspect Ratio Letterbox	(SZP 3)
Aspect Ratio Native	(SZP 1)
Auto Image Normal	(AIM 0)
Auto Image Wide	(AIM 1)
Brightness 10	(BRT 10)
Brightness 100	(BRT 100)
Contrast 10	(CON 10)
Contrast 100	(CON 10)
Freeze Off	(FRZ 0)
Freeze On	(FRZ 1)
Input 3G-SDI	(SIN+MAIN 7)
Input BNC	(SIN+MAIN 2)
Input Card Reader	(SIN+MAIN 11)
Input CVBS	(SIN+MAIN 9)
Input DisplayPort	(SIN+MAIN 6)
Input DVI-D	(SIN+MAIN 5)
Input HDBaseT	(SIN+MAIN 8)
Input HDMI 1	(SIN+MAIN 3)
Input HDMI 2	(SIN+MAIN 4)
Input Mini USB	(SIN+MAIN 12)
Input Presenter	(SIN+MAIN 10)
Input VGA	(SIN+MAIN 1)
Keypad 0	(KEY 36)
Keypad 1	(KEY 26)
Keypad 2	(KEY 27)
Keypad 3	(KEY 28)
Keypad 4	(KEY 29)
Keypad 5	(KEY 30)
Keypad 6	(KEY 31)
Keypad 7	(KEY 32)
Keypad 8	(KEY 33)
Keypad 9	(KEY 34)
Lamp Mode Constant Intensity	(LPM 1)
Lamp Mode Constant Power	(LPM 0)
Lamp Mode ECO 1	(LPM 2)
Lamp Mode ECO 2	(LPM 3)
Lamp Mode Rental Mode	(LPM 4)
Menu Navigation Down	(KEY 42)
Menu Navigation Enter	(KEY 40)
Menu Navigation Exit	(KEY 20)

Menu Navigation Left	(KEY 39)
Menu Navigation Menu	(KEY 19)
Menu Navigation Right	(KEY 41)
Menu Navigation Up	(KEY 38)
On Screen Display Off	(OSD 0)
On Screen Display On	(OSD 1)
PIP Input 3G-SDI	(SIN+PIP 7)
PIP Input BNC	(SIN+PIP 2)
PIP Input Card Reader	(SIN+PIP 11)
PIP Input CVBS	(SIN+PIP 9)
PIP Input DisplayPort	(SIN+PIP 6)
PIP Input DVI-D	(SIN+PIP 5)
PIP Input HDBaseT	(SIN+PIP 8)
PIP Input HDMI 1	(SIN+PIP 3)
PIP Input HDMI 2	(SIN+PIP 4)
PIP Input Mini USB	(SIN+PIP 12)
PIP Input Presenter	(SIN+PIP 10)
PIP Input VGA	(SIN+PIP 1)
PIP Off	(PIP 0)
PIP On	(PIP 1)
PIP Position Bottom Center	(PPP 3)
PIP Position Bottom Left	(PPP 5)
PIP Position Bottom Right	(PPP 4)
PIP Position Left Vertical Center	(PPP 0)
PIP Position Right Vertical Center	(PPP 2)
PIP Position Top Center	(PPP 1)
PIP Position Top Left	(PPP 6)
PIP Position Top Right	(PPP 7)
PIP Size Large	(PHS 2)
PIP Size Medium	(PHS 1)
PIP Size Small	(PHS 0)
PIP Swap None	(PPS)
Power Off	(PWR 0)
Power On	(PWR 1)

Appendix B. Query Commands

Aspect Ratio	(SZP?)
Brightness	(BRT?)
Contrast	(CON?)
Freeze	(FRZ?)
Input	(SIN+MAIN?)
Lamp Mode	(LPM?)
Lamp Usage	(LIF+LSHS?)
On Screen Display	(OSD?)
Operation Hours	(LIF+LPHS?)
PIP	(PIP?)
PIP Input	(SIN+PIP?)
PIP Position	(PPP?)
PIP Size	(PHS?)
Power	(PWR?)